Experiment Tracking and Model Registry using mlflow

Tuesday, January 27, 2026 6:00 PM

Recap: ML - Model building pipe

1 - 9 steps.

Load - EDA - prepr - Model - prepr - Eval
↳ Deploy.

iris - dataset

→ potential prob?

⇒ Generalization

over fitting and underfitting.

To Detect Model overfitting | underfitting

Cross validation

Divide train data

n parts

n-1 parts Train

rem-part Test

n x Train perf

n x Valid perf

X we shouldn't Compare

X Traindata perf ≈ Test data perf

Repetitively

Data leakage

Mean Train perf ≈ Mean valid perf

reasonably high

Diff b/w them < 5%

Config used for CV

Good Conf

Generalized Conf

Neither simple Nor complex

Both perf are very poor

Underfit

Train perf >> valid perf

Overfit.

Dataset; 100 obs.

ex: KNN.

Dataset; 100 obs.

ex: KNN.

Config → n_neigh = 100 (sample)

→ low Train perf ✓ Valid perf in CV.

→ Under fit

hyp par

Config → n_neigh = 1 (complex)

→ high Train perf low Valid perf

↓

Overfit

Found: either overfit or under fit

→ Need to experiment with another Config.

↓

Neither overfit nor underfit

↙

To perf → Manually

↓

Tedious

An automated way to expt with diff Config of model

Hyper param Tuning.

→ Model Config → Neither overfit Nor underfit

→ Sklearn

→ optuna ✓

parameter ⇒ learn't during training optuna

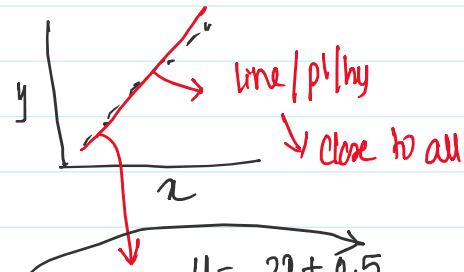
Hyper param. → Improves Capability of model ML flow

param: Lin Reg

→ $y = w_1x_1 + w_2x_2 + \dots + w_nx_n + w_0$

Dataset

x	y



$y = 2x + 0.5$

$y = w_1 x + w_0$

$w_0 = 0.5$, $w_1 = 2$ } obtained by Train → parameters → DL

Trainable params.

DT → If-else-then → Built based on data during Train

(If-else-then: param)

n_neigh = 1 → Complex = overfit

n_neigh = 5 → Generalized model

n_neigh = 100 → Simple = underfit

Imp

Cap model → hyper param.

ml flow → Track data gen during expt → Saved → Visualizing / Analyzing; Make dec abt model

Expt → check differ Config

Optuna → Check all | better Config of model → To find the generalized Config → Further Deployed using CI.

Hyp param Tuning → Different ways to shopper for Best Config

1) Grid search → evaluate a set of Combination

KNN → n_neigh, weights, p.

used in Train time

KNN → n_neigh , weights , p.

Fix	1 to # obs	'U'	1
3 to 20.		'D'	2
			3.

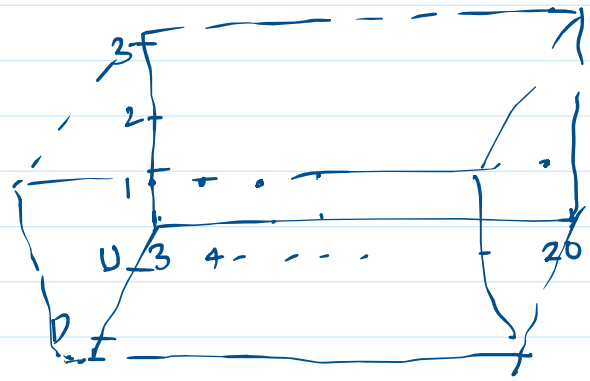
Used → Train time → Small

Grid search

ML algo.

Random Search

→ Out of all the planned ranges



Training time is Large

e.g.: 50 runs

→ pick 50 runs / Config rand

→ find out Best run

Sequential search / Bayesian search

→ optuna → TPE

Train time ↑ large

→ Finds the Config wisely

Bayesian rule